

Class Activity

1. Hospital Patient Admission Analysis

R Program

```
library(ggplot2)
library(dplyr)
library(treemap)

# Dataset
hospital <- data.frame(
  Department = c("Cardiology","Neurology","Pediatrics",
                 "Orthopedics","Emergency"),
  Admissions = c(320,250,410,280,520)
)

# Expanded data for boxplot and histogram
set.seed(100)
patient_data <- data.frame(
  Department = rep(hospital$Department, each=50),
  Patients = c(rpois(50,20),
              rpois(50,15),
              rpois(50,25),
              rpois(50,18),
              rpois(50,30))
)

# Bar Chart
ggplot(hospital,aes(x=Department,y=Admissions,fill=Department))+
  geom_bar(stat="identity")+
  ggtitle("Hospital Admissions by Department")

# Stacked Bar Chart
ggplot(hospital,aes(x="Admissions",y=Admissions,fill=Department))+
  geom_bar(stat="identity")+
```

```

ggtitle("Stacked Bar Chart")

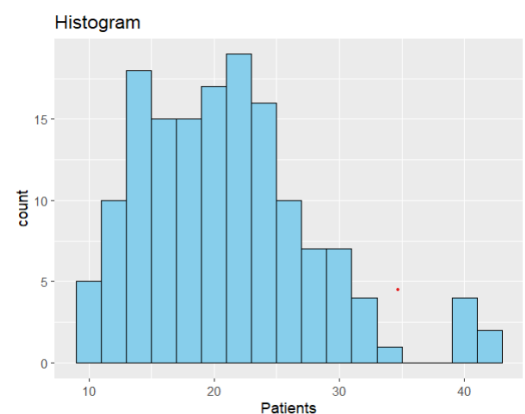
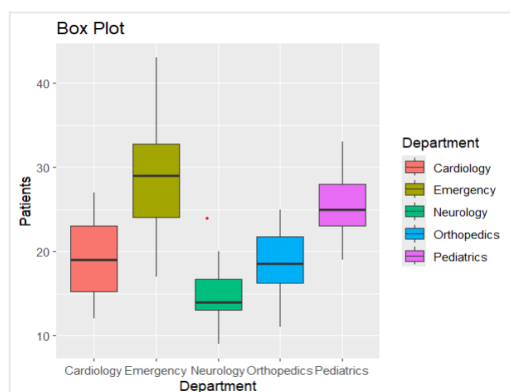
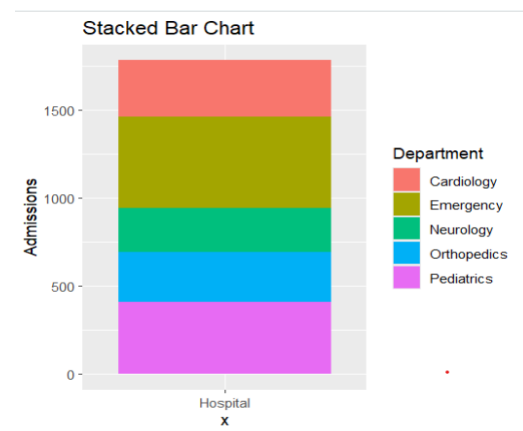
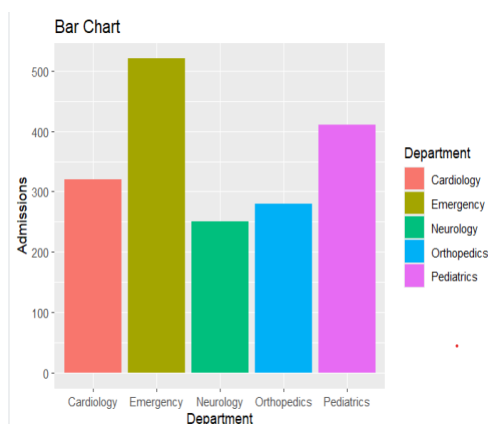
# Treemap
treemap(hospital,
        index="Department",
        vSize="Admissions",
        title="Department Workload")

# Box Plot
ggplot(patient_data,aes(x=Department,y=Patients,fill=Department))+
  geom_boxplot()+
  ggtitle("Admission Variability")

# Histogram
ggplot(patient_data,aes(Patients))+
  geom_histogram(fill="skyblue",bins=10)+
  ggtitle("Patient Distribution")

```

Output



2. E-Commerce Customer Spending Behaviour

R Program

```
library(ggplot2)

set.seed(101)

customers <- data.frame(
  Spending=round(rnorm(200,5000,1500)),
  Orders=sample(1:20,200,replace=TRUE),
  Category=sample(c("Electronics","Fashion","Home","Sports"),200,replace=TRUE)
)

# Histogram
ggplot(customers,aes(Spending))+
  geom_histogram(fill="orange",bins=20)+
  ggtitle("Customer Spending Histogram")

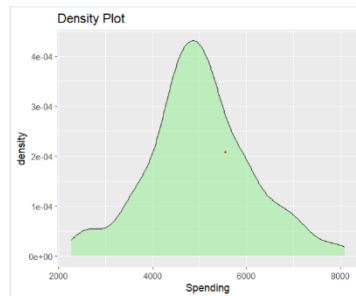
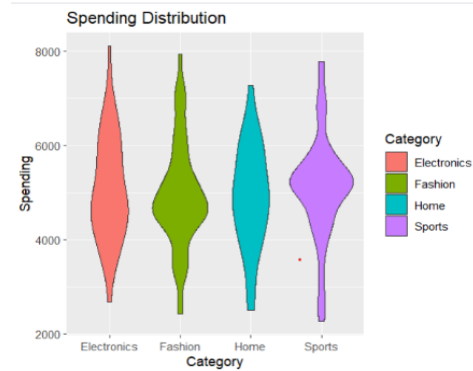
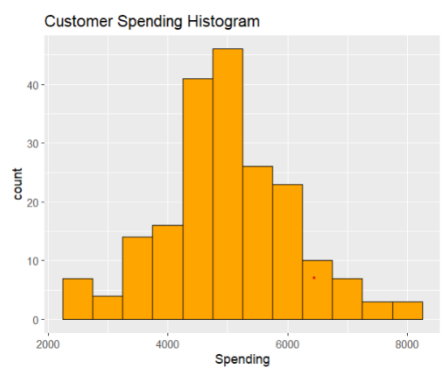
# Violin Plot
ggplot(customers,aes(Category,Spending,fill=Category))+
  geom_violin()+
  ggtitle("Spending by Category")

# Box Plot
ggplot(customers,aes(Category,Spending,fill=Category))+
  geom_boxplot()+
  ggtitle("Customer Spending Boxplot")

# Density Plot
ggplot(customers,aes(Spending))+
  geom_density(fill="lightgreen",alpha=0.5)+
  ggtitle("Density Plot")

# Bar Chart
ggplot(customers,aes(Category,fill=Category))+
  geom_bar()+
  ggtitle("Orders by Category")
```

Output



3. University Examination Performance Dashboard

R Program

```
library(ggplot2)
```

```
set.seed(102)
```

```
students <- data.frame(
```

```
  Department=sample(c("CSE","AI","IT","ECE"),200,replace=TRUE),
```

```
  Marks=round(rnorm(200,75,10))
```

```
)# Box Plot
```

```
ggplot(students,aes(Department,Marks,fill=Department))+
```

```
  geom_boxplot()+
```

```
  ggtitle("Department-wise Marks")
```

```
# Histogram
```

```
ggplot(students,aes(Marks))+
```

```
  geom_histogram(fill="steelblue",bins=15)+
```

```

ggtitle("Marks Distribution")

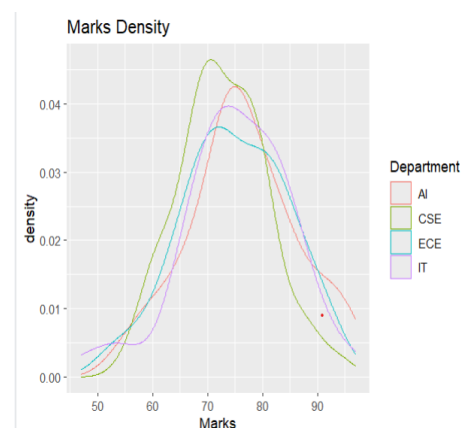
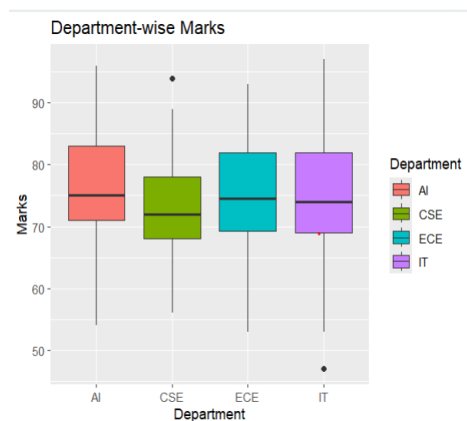
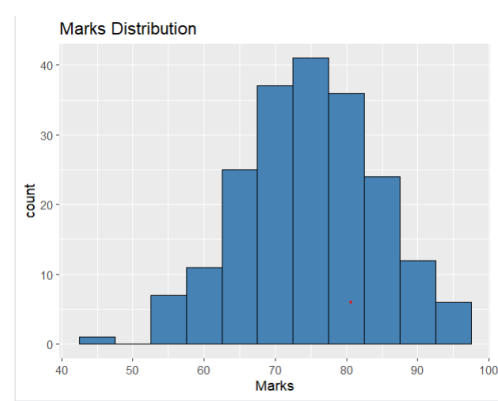
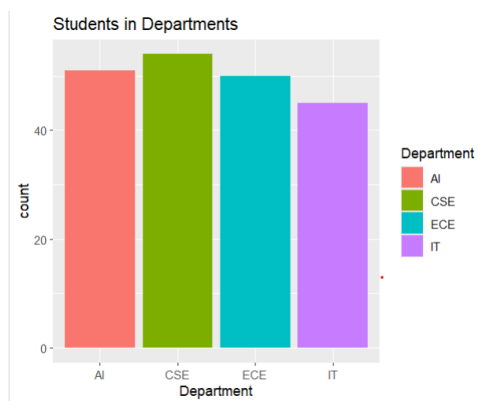
# Bar Chart
ggplot(students,aes(Department,fill=Department))+
  geom_bar()+
  ggtitle("Students by Department")

# Density Plot
ggplot(students,aes(Marks,color=Department))+
  geom_density()+
  ggtitle("Marks Density")

# Plotly Scatter
plot_ly(students,
  x=~Department,
  y=~Marks,
  color=~Department,
  type="scatter",
  mode="markers")

```

Output



4. Climate and Rainfall Distribution Study

R Program

```
library(ggplot2)

climate <- data.frame(
  City=rep(c("Chennai","Bangalore","Hyderabad"),each=12),
  Month=rep(month.abb,3),
  Rainfall=sample(40:300,36),
  Temperature=sample(22:38,36),
  Humidity=sample(55:95,36)
)

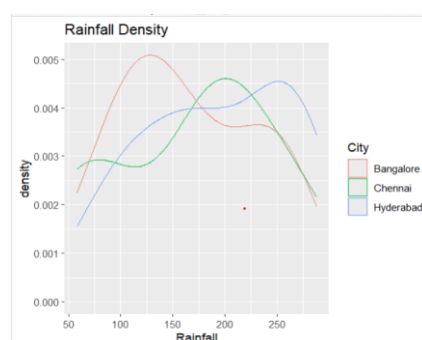
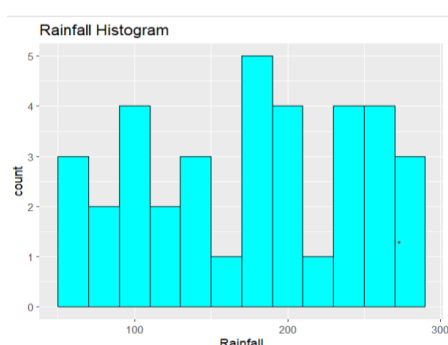
# Histogram
ggplot(climate,aes(Rainfall))+
  geom_histogram(fill="cyan",bins=12)+
  ggtitle("Rainfall Histogram")

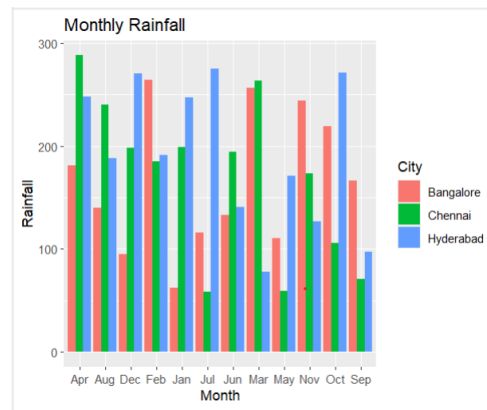
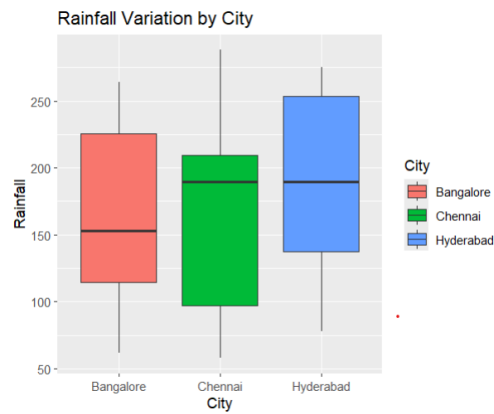
# Density Plot
ggplot(climate,aes(Rainfall,color=City))+
  geom_density()+
  ggtitle("Rainfall Density")

# Box Plot
ggplot(climate,aes(City,Rainfall,fill=City))+
  geom_boxplot()+
  ggtitle("Rainfall by City")

# Bar Chart
ggplot(climate,aes(Month,Rainfall,fill=City))+
  geom_bar(stat="identity",position="dodge")+
  ggtitle("Monthly Rainfall")
```

Output





5. Bank Loan Risk Assessment

R Program

```
library(ggplot2)
set.seed(103)
bank <- data.frame(
  Loan=round(rnorm(200,400000,100000)),
  Income=round(rnorm(200,60000,15000)),
  CreditScore=sample(500:900,200,replace=TRUE),
  Status=sample(c("Low Risk","Medium Risk","High Risk"),
    200,replace=TRUE)
)
# Histogram
ggplot(bank,aes(Loan))+
  geom_histogram(fill="gold",bins=15)+
  ggtitle("Loan Distribution")
# Box Plot
ggplot(bank,aes(Status,Loan,fill=Status))+
  geom_boxplot()+
  ggtitle("Loan Amount by Risk")
# Scatter Plot
ggplot(bank,aes(CreditScore,Loan,color=Status))+
  geom_point(size=2)+
  ggtitle("Credit Score vs Loan")
```

Bar Chart

```
ggplot(bank,aes(Status,fill=Status))+  
  geom_bar()+  
  ggtitle("Customer Risk Levels")
```

Density Plot

```
ggplot(bank,aes(Loan,fill=Status))+  
  geom_density(alpha=0.5)+  
  ggtitle("Loan Density by Risk")
```

Output

